Cambridge (CIE) A Level Maths: Probability & Statistics 1

Your notes

Statistical Measures

Contents

- Basic Statistical Measures
- Frequency Tables
- Standard Deviation & Variance
- Coding Data

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Your notes

Mean, Mode, Median

What are mean, median and mode?

- Mean, median and mode are measures of **location**
  - A measure of location gives information about where data is in the number system
  - Mean, median and mode are **measures of central tendency**
  - They describe where the centre of the data is
- They are all types of **averages**
  - In Statistics it is important to be specific about which average you are referring to

How are mean, median and mode calculated?

- You should already be familiar with finding the mean, median and mode from raw, ungrouped data
- The **mode** is the value that occurs **most often** in a data set
  - In a frequency table the **group** or **class** that occurs most often will be referred to as the **modal class**
  - A data set with more than one mode is **bimodal**
- The **median** is the **middle** value when the data is in **order of size**
  - If there are two values in the middle of the data set, the median is the **midpoint** of the two values
  - If finding median from a **frequency table** find the cumulative frequency first and find the group or class where the middle value will lie
- The **mean** is the **sum** of all the values **divided by the number of values** in the data set

What are summary statistics and their notation?

- Summary statistics are information that summarises a set of data values
- For n items in a data set:
  - The sum of the data is represented by

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + \ldots + x_n$$

- This is usually written $\sum x$ and reads as 'sigma x'
- The mean of the data is represented by

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$$\bar{x} = \frac{x_1 + x_2 + \ldots + x_n}{n} = \frac{\Sigma x}{n}$$

- This reads as 'x bar'
- You will come across more summary statistics later in the course

How do we choose the best measure of central tendency?

- It is often better to use one of the averages over the others, depending on the data set
  - It's a good idea to be aware of the advantages and disadvantages of the use of each average
- The mean uses all of the data values, this is good for a large data set where all of the values are close together, but also means that the mean can be affected by extreme values
- The median is not affected by very high or low values so is a good average to use in data sets with extreme values
- The mode is very useful in a lot of practical situations, however often there may be more than one mode, no mode or even a mode that is nowhere near the middle of the data set

Worked Example

For the data set given below, find the mode, median and mean.

23 19 14 28 27 19

23 19 14 28 27 19

Mode = most common number = 19

Mode = 19

To find the median the data must be in order first

14 19 19 23 27 28

Median is half way between the two middle numbers.

$$\text{Median} = \frac{19 + 23}{2} = 21$$

Median = 21

The mean is the sum of the numbers divided by how many there are.

$$\text{Mean} = \frac{\Sigma x}{n} = \frac{14 + 19 + 19 + 23 + 27 + 28}{6} = \frac{130}{6}$$

Mean = 21.7 (3.s.f.)

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Quartiles & Range

Your notes

What are quartiles?

- **Quartiles and percentiles** are measures of location
- Quartiles divide a population or data set into **four equal sections**
  - The **lower quartile,** Q_1 splits the lowest 25% from the highest 75%
  - The **median,** Q_2 is the value that is 50% of the way through the data
  - The **upper quartile,** Q_3 splits the lowest 75% from the highest 25%

Your notes

How are quartiles calculated?

- For a data set of size, n,
  - To find the lower quartile, calculate $\frac{n+1}{4}$
    - If $\frac{n+1}{4}$ is an integer then the lower quartile is the corresponding value
    - If $\frac{n+1}{4}$ is not an integer then the lower quartile is the midpoint of the two values corresponding to the integers either side of $\frac{n+1}{4}$
  - To find the upper quartile, calculate $\frac{3(n+1)}{4}$
    - If $\frac{3(n+1)}{4}$ is an integer then the upper quartile is the corresponding value
    - If $\frac{3(n+1)}{4}$ is not an integer then the upper quartile is the midpoint of the two values corresponding to the integers either side of $\frac{3(n+1)}{4}$
- The median can be found in the same way using $\frac{n+1}{2}$

What are the range and interquartile range?

- The **range** and **interquartile range** are both **measures of spread**
- A measure of spread gives information about how spread out the data set is
- The **range** is the difference between the **largest** and **smallest** values in the data set
  - All data points in the set will be included in the range, including **extreme values**
- The **interquartile range** is the difference between the **upper quartile** and the **lower quartile**
  - Only the middle 50% of the data is included in the interquartile range

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- It is not affected by extreme values
- The units for range and interquartile range are the same as the units for the original data

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Worked Example

Find the range and interquartile range for the data set given below

43 29 70 31 84 56 17 53.

Always start by putting the values in order

17 29 31 43 53 56 70 84

Range is the difference between the biggest and smallest

$$\text{Range} = 84 - 17 = 67$$

$$\text{Range} = 67$$

Interquartile range = upper quartile − lower quartile

$$n = 8$$

$\frac{8+1}{4} = 2.25$ so lower quartile is midpoint of 2nd and 3rd values

$$Q_1 = \frac{29+31}{2} = 30$$

$\frac{3(8+1)}{4} = 6.75$ so upper quartile is midpoint of 6th and 7th values

$$Q_3 = \frac{56+70}{2} = 63$$

This is easier to see by looking at the data directly

17 29 31 43 53 56 70 84

Q_1 Q_2 Q_3

$$\text{Interquartile range} = Q_3 - Q_1 = 63 - 30 = 33$$

$$\text{IQR} = 33$$

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Examiner Tips and Tricks

- Be aware of the difference between averages and ranges, especially when answering contextual questions asking you to describe or compare data. Remember, averages give an indication of where the data are whilst range gives an indication of how varied the data are.

Your notes

Ungrouped data

A frequency table for **ungrouped data** shows the frequency of **individual data values**

Why use a frequency table for ungrouped data?

- When collecting large amounts of raw data, it is quicker to use a tally system and then collate the results into a frequency table
- Ungrouped frequency tables are normally used for **numerical, discrete data**
- Organising data into a frequency table makes it easier to work with
  - Calculating averages, ranges and summary statistics can be done much quicker from a frequency table than from raw data
- It gives a clear **pattern** of the data
  - It is easy to quickly see where most of the data are and to see extreme values
- A frequency table for ungrouped data keeps all of the original data values
  - It is still possible to calculate the actual averages, ranges and summary statistics

How are mean, median and mode calculated from an ungrouped frequency table?

- You should already be familiar with finding the mean, median and mode from raw, ungrouped data
- The **mode** is the value that occurs **most often** in a data set
  - In an ungrouped frequency table the **data value** with the **highest frequency** will be the mode
- The **median** is the **middle** value when the data is in **order of size**
  - To find the median from an ungrouped frequency table, add the frequencies together until you reach the value that is half of the total
  - For a data set of values, calculate $\frac{n+1}{2}$ and the median will be whichever data value corresponds with this frequency
- The **mean** is the **sum** of all the values **divided by the number of values** in the data set
  - To find the mean, multiply each data value by its corresponding frequency, find the sum of these values and then divide by the total frequency
  - The notation for the sum of the values is Σxf
  - The formula for the mean from a frequency table is $\frac{\Sigma xf}{\Sigma f}$

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Worked Example

The frequency table below gives information about the shoe size of a group of year 12 students.

Shoe Size	37	37.5	38	38.5	39	39.5	40	40.5	41
Frequency	3	3	12	5	16	9	7	4	1

(i) Write down the modal shoe size.

(ii) Explain how you know that the median shoe size is 39.

(iii) Calculate the mean shoe size.

(i) Modal shoe size is the most common shoe size.

the highest frequency = 16

Modal shoe size = 39

(ii) To find the median we first need to know the total number of students:

$f = 3 + 3 + 12 + 5 + 16 + 9 + 7 + 4 + 1 = 60$

The data is discrete so we use $\frac{n+1}{2}$ to find which value the media will be.

$\frac{60+1}{2} = 30.5^{th}$ value

Use the cumulative frequencies to find which data value is half way between the 30th and the 31st.

Shoe size	37	37.5	38	38.5	39	39.5	40	40.5	41
Frequency	3	3	12	5	16	9	7	4	1
Cumulative frequency	3	6	18	23	39				

+3 +12 +5 +16

We are looking for the 30.5^{th} value so we don't need to go further

The median shoe size is half way between the 30th and 31st shoe size which are both 39, therefore median = 39.

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(iii)

Shoe size, x	37	37.5	38	38.5	39	39.5	40	40.5	41
Frequency, f	3	3	12	5	16	9	7	4	1
xf	111	112.5	456	192.5	624	355.5	280	162	41

$$\text{mean} = \bar{x} = \frac{\Sigma xf}{\Sigma f} = \frac{111 + 112.5 + 456 + \ldots + 162 + 41}{60} = \frac{2334.5}{60}$$

Mean = 38.9 (3.s.f.)

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Your notes

Examiner Tips and Tricks

- Use common sense when checking your answers, is your mean within the range of the data? Does is seem right? A mean of 140 for example could not be correct if the data is about ages of students taking an exam at university.

Grouped data

A frequency table for **grouped data** is usually used for large amounts of **continuous data.** They shows the frequency of **data values** that are within a particular **group** or **class.**

What are the advantages and disadvantages of grouping data?

- Grouping data is especially useful when data can take a large range of different values
- Trends and patterns can be easily spotted when data has been grouped
- Calculations are much quicker with data that has been grouped
- It is important to be aware, however, that grouped frequency tables also have some negatives
  - The actual data values are lost when data is grouped
  - It is only possible to calculate estimated averages, ranges and summary statistics

What notation is used for grouped frequency tables?

- When grouping data, it is important to be clear about which **group** or **class** any data value should be entered into
  - A group entry of 10 – 20 followed by 20 – 30 would be unclear because the number 20 could be entered into both groups
    - If the data are discrete, this could be written as 10 – 19 and 20 – 29
    - For continuous data, this could be changed to 10 – and 20 –
    - This would most likely be represented as $10 \leq x < 20$ and $20 \leq x < 30$

- Most commonly inequalities are used to group continuous data as they leave no ambiguity
- If the data are continuous, always check that there are no gaps between upper boundary of a class and the lower boundary of the next class
  - If there are gaps you will need to close these gaps by changing the boundaries before carrying out any calculations
    - For example, the groups $10 \leq x \leq 19$ followed by $20 \leq x \leq 29$ will become $9.5 \leq x < 19.5$ and $19.5 \leq x < 29.5$
    - Check the inequality signs carefully
- Be careful when deciding what category the data falls into, taking the group $10 \leq x \leq 19$ for example,
  - if the data had been rounded it would take the form $9.5 \leq x < 19.5$ as described above
  - if the data had been **truncated** however, then the boundaries would become $10 \leq x < 20$
    - this is most likely to happen with age, which is technically a continuous variable due to being able to take any, however we would usually consider age by counting years

How do I find averages from grouped frequency tables?

- Instead of finding the **mode**, when working with a grouped frequency table we instead find the **modal class**
  - This will be the **class** (group) with the **greatest frequency**
- It is only possible to calculate an **estimate** for the **mean** and the **median** from a grouped frequency table
- Calculating the **estimated mean** is the same as for ungrouped frequency tables, however you will need to find the **midpoints** first
  - the midpoint is the mean of the upper and lower values in the class boundaries
  - multiply the midpoint by its corresponding frequency, find the sum of these values and then divide by n
- Calculating the **estimated median** is more complicated and questions will only ask for the class that the estimated median would be in

Worked Example

The table below shows the heights in cm of a group of year 12 students.

Your notes

Height, h	Frequency
$150 \le h < 155$	3
$155 \le h < 160$	5
$160 \le h < 165$	9
$165 \le h < 170$	7
$170 \le h < 175$	1

(i) Write down the class that a student of exactly 160 cm should be added to.

(ii) Write down the modal class.

(iii) Calculate the estimated mean height.

(i) The two classes with 160 in are:

$155 \leqslant h < 160$ Not equal to

$160 \leqslant h < 165$ Or equal to

160 must go into the $160 \leqslant h < 165$ class

(ii) The modal class is the group with the greatest frequency.

Modal class = $160 \leqslant h < 165$

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(iii) Add the midpoint to the table and then multiply it by the frequency.

Height, h, cm	Midpoint, x	Frequency, f	xf
$150 \leqslant h < 155$	152.5	3	152.5 x 3 = 457.5
$155 \leqslant h < 160$	157.5	5	157.5 x 5 = 787.5
$160 \leqslant h < 165$	162.5	9	162.5 x 9 = 1462.5
$165 \leqslant h < 170$	167.5	7	167.5 x 7 = 1172.5
$170 \leqslant h < 175$	172.5	1	172.5 x 1 = 172.5

$\Sigma fx = 457.5 + 787.5 + 1462.5 + 1172.5 + 172.5 = 4052.5$

$\Sigma f = 3 + 5 + 9 + 7 + 1 = 25$

$$\text{Estimated mean} = \frac{\Sigma fx}{\Sigma f} = \frac{4052.5}{25} = 162.1$$

Estimated mean height = 162 cm (3.s.f.)

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Your notes

Your notes

Examiner Tips and Tricks

- There can be a lot of calculations when working with grouped frequency tables, be extra careful when using your calculators as it is easy to make small errors with these questions. Use the table and add information to the table as you go as there will be marks available for showing work within the methods.

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Your notes

Standard Deviation & Variance

The **variance** is another measure for the spread of the data, it measures the variability from the mean of the data.

What is the variance and the standard deviation?

- The variance is a statistic that tells us how **varied** a set of data is
  - Data that is more spread out will have a greater variance
  - Data that is consistent and close together will have a smaller variance
- The **standard deviation** is the square root of the variance
- The symbol for the population standard deviation is the lowercase Greek letter sigma, **σ** and for variance is sigma squared, $\boldsymbol{\sigma}^2$
- Standard deviation and variance are used interchangeably within this course so make sure you look out for which one a question shows or asks for

How are the variance and standard deviation calculated?

- There is more than one formula that can be used for calculating the variance, and you should choose the most useful one
- For a set of n values $x_1, x_2, \dots, x_i, \dots, x_n$ the variance is the sum of the squares of the deviations from the mean, divided by the frequency
  - Variance = $\dfrac{\Sigma(x-\bar{x})^2}{n}$
    - This formula can be time consuming and therefore is rarely used in this statistics course
- A second, **easier to use version of the variance** is:
  - Variance = $\dfrac{\Sigma x^2}{n} - \bar{x}^2$
  - This version is easier to work with and should be used in most instances
  - Variance can also be written in other ways

$$\text{Variance} = \sigma^2 = \frac{\Sigma(x-\bar{x})^2}{n} = \frac{1}{n}\left(\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right) = \frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2 = \frac{\Sigma x^2}{n} - \bar{x}^2$$

- An easy way to remember this is to think of it as '*the sum of x squared over n minus the sum of the mean squared*'
- Most calculators can be used to find summary statistic such as the standard deviation and variance fairly quickly, practice finding it on yours

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- The **standard deviation** is the square root of the variance
  - Standard deviation = $\sigma = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n}} = \sqrt{\frac{\Sigma x^2}{n} - \bar{x}^2}$
    - Makes sure you know how to find these formulae in the formula booklet and are familiar with the version given
- The units for standard deviation are the same as the units for the data and the units for variance are the same as the units for the data but squared

Your notes

How are the variance and standard deviation calculated from a frequency table?

- The method for finding the variance from a frequency table is similar to that of the mean
  - If calculating from a grouped frequency table, find the midpoints, x, first
  - Multiply each x value by its corresponding frequency and use these values within the formulae
  - The formulae will become
    - Variance = $\sigma^2 = \frac{\Sigma(x-\bar{x})^2 f}{\Sigma f} = \frac{\Sigma x^2 f}{\Sigma f} - \left(\frac{\Sigma xf}{\Sigma f}\right)^2$
    - Standard deviation = $\sigma = \sqrt{\frac{\Sigma(x-\bar{x})^2 f}{\Sigma f}} = \sqrt{\frac{\Sigma x^2 f}{\Sigma f} - \left(\frac{\Sigma xf}{\Sigma f}\right)^2}$

Worked Example

Phoom recorded the length of time, t, it took him, in minutes, to answer a selection of further calculus exam questions. The data is summarised in the table below.

Time (minutes)	Frequency
$2 \le t < 4$	1
$4 \le t < 6$	4
$6 \le t < 8$	7
$8 \le t < 10$	5
$10 \le t < 12$	2
$12 \le t < 20$	2

(i) Calculate an estimate of the variance of the time taken to complete a question, include units with your answer.

(ii) Write down an estimate of the standard deviation of the time taken to complete a question, include units with your answer.

Your notes

(i)

Find the midpoint, h, of each group:

Time, (mins)	Midpoint, t	t^2	Frequency, f	ft	ft^2
$2 \leqslant t < 4$	3	9	1	3	9
$4 \leqslant t < 6$	5	25	4	20	100
$6 \leqslant t < 8$	7	49	7	49	343
$8 \leqslant t < 10$	9	81	5	45	405
$10 \leqslant t < 12$	11	121	2	22	242
$12 \leqslant t < 20$	16	256	2	32	512

This group has a different width to the others.

Find the summary statistics:

Your calculator can be used in 'STAT' mode to find these values

$\Sigma ft = 171 \qquad \Sigma ft^2 = 1611 \qquad \Sigma f = 21$

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(i) Substitute into the formula for variance

$$\sigma^2 = \frac{\Sigma ft^2}{\Sigma f} - \left(\frac{\Sigma ft}{\Sigma f}\right)^2 = \frac{1611}{21} - \left(\frac{171}{21}\right)^2$$

$$= \frac{510}{49} = 10.408$$

Estimated variance = 10.4 minutes^2 (3.s.f.)

(ii) The standard deviation is the square root of the variance

$$\sigma = \sqrt{\frac{510}{49}} = \frac{\sqrt{510}}{7} = 3.2261...$$

Estimated standard deviation = 3.23 minutes (3.s.f.)

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Examiner Tips and Tricks

- Look out for whether a question gives or asks for the standard deviation or variance, especially if the question is using sigma notation.
- Choose which formula to use wisely, most of the time the summary statistics will be given so only one of the formulae will be possible. On the rare occasion that

you are asked to calculate directly from a table think carefully about which version of the formula is quickest and easiest to use. It will almost always be the second version given in this revision note.

Your notes

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Your notes

Coding

Sometimes data needs to be coded for further use with calculations. This is particularly useful with data that deals with very small or very large numbers, or with data that needs to be classified for research purposes.

What is coding?

- Coding is a way of simplifying data to make it easier to work with
- The coding must be carried out on all values within the data set and will normally be done using a given formula
- Coding can be carried out in a number of ways:
  - Adding or subtracting a constant to each data value
  - Multiplying or dividing each data value by a constant
  - A combination of both of the above
- You may occasionally see coding called an **assumed mean**
  - This refers to the value chosen to subtract from the original data
  - It is usually chosen as an easy value, similar to an estimate of the true mean

How are statistical calculations carried out with coded data?

- If you know the **mean** or **standard deviation** of the original data it is possible to find the mean or standard deviation of the coded data and vice versa
- It is important to remember what the **mean** and **standard deviation** actually tell us about the data to understand how coding calculations work
  - The mean is a measure of **location**, changing the data set in any way will cause the mean to change in the same way
  - The **standard deviation** is a measure of spread, adding or subtracting a constant to every value within the data set will **not change** the standard deviation of the data set
    - Multiplying or dividing every value within the data set by a constant will change the standard deviation by the **modulus** of the constant
    - If the data were coded by multiplying or dividing by a **negative**, the standard deviation will change by the equivalent positive value
- Anytime calculations are carried out on data that has been coded,
  - The **original mean** can be found by solving the equation to reverse the coding
  - For example, if the data, x, was coded using the formula

$$y = ax + b$$

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Then the mean of the coded data, $\bar{y}$ would be

$$\bar{y} = a\bar{x} + b$$

The original mean, $\bar{x}$, will be

$$\bar{x} = \frac{\bar{y} - b}{a}$$

- The **original standard deviation**
  - Will be the same as the coded standard deviation if the data was coded by adding or subtracting a constant only
  - Can be found by reversing the coding if the data was coded by multiplying or dividing by a constant only
  - If the data was coded by a combination of both then only the multiplying or dividing will need to be reversed to find the original standard deviation
- For example, if the data, x, was coded using the formula

$$y = ax + b$$

Then the standard deviation of the coded data, σ_y would be

$$\sigma_y = |a|\sigma_x$$

The original standard deviation, σ_x, will be

$$\sigma_x = \frac{\sigma_y}{|a|}$$

What will summary statistics and formulae look like with an assumed mean?

- If an **assumed mean**, *a*, has been subtracted from all data values, *x*, then the summary statistics for the coded data will be
  - The sum of the coded data $\Sigma(x - a)$
  - The sum of the squares of the coded data $\Sigma(x - a)^2$
    - Be careful not to mix this up with the square of the sum of the coded data $(\Sigma(x - a))^2$
  - If an **assumed mean,** *a*, has been subtracted from all data values, *x*, then the formulae for the mean and standard deviation for the coded data will be

$$\overline{(x - a)} = \frac{\Sigma(x - a)}{n}$$

Your notes

$$\sigma_{x-a} = \sqrt{\frac{\Sigma(x-a)^2}{n} - \left(\frac{\Sigma(x-a)}{n}\right)^2}$$

Your notes

- Most questions will give you either the summary statistics or the mean of the coded data and expect you to work with these formulae to find original information about the data

Worked Example

A coffee machine is set to dispense 150 ml of coffee per cup.

In a random sample of 20 cups of coffee $\Sigma(c-150) = -16$, where c ml is the volume of coffee in a cup.

(a) Find the mean volume in a cup of coffee.

(b) Given that $\Sigma(c-150)^2 = 112$, find the standard deviation of the sampled cups of coffee.

a) Use the assumed mean to find the coded mean:

$$\overline{c-150} = \frac{\Sigma(c-150)}{n} = \frac{-16}{20} = -0.8$$

Solve to find the true mean:

$$\bar{c} = \overline{c-150} + 150 = -0.8 + 150 = 149.2$$

True mean = 149.2 ml = 149 ml (3 sf)

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b) Use the assumed mean within the formula for the standard deviation:

$$\sigma_{c-150} = \sqrt{\frac{\Sigma(c-150)^2}{n} - \overline{(c-150)}^2}$$

$$= \sqrt{\frac{112}{20} - (-0.8)^2}$$

$$= \sqrt{4.96}$$

$$= 2.227...$$

No need to undo any coding as no multiplication or division was carried out

Standard deviation = 2.23 ml (3 sf)

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Examiner Tips and Tricks

- Be careful when using the formulae for the mean and standard deviation with coded summary statistics, you must make sure that you use the summary statistics consistently throughout. For example, if you use the sum of the coded data squared in the formula for the standard deviation, you must subtract the square of the coded mean.

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